AI Learning & Application Workflow

This workflow is designed to help you (Shreyansh) become a master in Artificial Intelligence, focusing not on jobs but on building AI systems smarter than existing models. The workflow is divided into learning stages and application stages.

1. Learning Path

1. Mathematics Foundation: Linear Algebra, Probability, Statistics, Calculus.
2. Programming Skills: Python (NumPy, Pandas, Matplotlib), OOP, data structures & algorithms.
3. Machine Learning: Supervised/Unsupervised Learning, Scikit-learn, model evaluation.
4. Deep Learning: Neural networks, CNNs, RNNs, Transformers (PyTorch/TensorFlow).
5. Natural Language Processing: Embeddings, attention, transformers, tokenization.
6. Open-source LLMs: LLaMA, Mistral, Falcon – how they work & fine-tuning.
7. Reinforcement Learning: RL basics, PPO, RLHF (Reinforcement Learning with Human Feedback).
8. Systems & Scaling: Distributed computing, GPUs, cloud, vector databases.
9. Ethics & Safety: Bias, interpretability, AI alignment concepts.

2. Application Path

1. Start with small projects: Spam classifier, image recognition, text summarizer.
2. Recreate GPT-like models in small scale (character-level language model).
3. Fine-tune open-source models (e.g., LLaMA) on your own datasets.
4. Build AI agents (autonomous task execution with tools & APIs).
5. Experiment with multi-modal AI (vision + text + audio).
6. Work on scalability: deploy AI on cloud, optimize inference.
7. Innovate: Try to design architectures beyond transformers (new research).

■ *Tip: Learn one stage at a time. Watch YouTube playlists, read blogs, and implement projects after each stage. Mastery comes from applying what you learn.*